



NATIONAL UNIVERSITY
of Computer & Emerging Sciences

SE Project Proposal

Project Name (Mindscribe)

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Problem Statement:

In the modern world, a large amount of information is stored and utilized by various people. Usually, they rely on bookmarks or some tools like notepad or note taking apps. Although note taking apps can serve a great deal of help, the real problem lies with efficient **utilization**.

The potential problems are:

- Ideas can be fragmented and lost between various notes
- There is always a margin of error when a user has to list out some points i.e leaving important details, contradicting points.
- Users often forget the existence of a particular note, leading to duplication of information

To solve these problems, we introduce **Mindscribe**, an AI and RAG powered application with the purpose of making your information storage and retrieval easier and efficient.

Problem Scope:

An object of information which is stored is called as a “**note**” in our project

The features it will include:

- Hybrid Searching: Searching and retrieval of notes by semantic processing or utilizing metadata
- Knowledge graph: A visual representation of notes based on similarity
- Critic: Evaluation of written notes
- AutoGeneration: Auto Creation of notes based on details provided by user prompts
- Summarization: Summary of all stored notes based on a topic provided by user query
- Memorization: Maintaining history of users created notes to improve experience
- Context radar: A list of stored notes based on note currently worked on

The application will be web-based only, with supporting voice and text inputs. Users must have a stable internet connection and is for personal use - not for enterprise level

Tools and Technologies:

Frontend:

React (Javascript) since it is a standard for frontend projects

Backend:

FASTApi (Python). Since most backend code will rely on AI based libraries with heavy support of Python.

AI Engine: Langchain and LangGraph

Database: PostgreSql for metadata and FAISS vector store for storing embeddings

Third party library: pgvectors(PostgreSql extension) for handling embeddings and metadata simultaneously

Review of Similar Projects:

There are several similar projects of our domain, but with certain limitations:

- **Notion:** It is used for managing documents and managing tasks, but it cannot link docs(notes) to each other based on similarity
- **Obsidian:** Personal knowledge management tool, but with manual linking of information, cannot relate automatically.
- **Grammarly:** A really famous application for fixing grammar and spelling.

Unique factor:

The things which makes our project a different is that we have approached a hybrid way of approaching this project such that:

- Mindscribe will automatically formulate links and link strength between notes instead of manual linking
- Instead of providing grammatical or spelling mistake fixed, it can also recommend proper structure of ideas based on the context
- It may be able to deal with images semantically
- We can summarize information of similar notes at once with a single action instead of surfacing through all of those notes